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Final Report

Robot Receptionist

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Abstract

Human receptionists face challenges with 24/7 availability, repetitive tasks and fatigue. This project aimed to develop an autonomous robot receptionist using the Pepper robot to address these challenges. An Agile development approach was maintained to allow for the constantly changing requirements. A Python 3 to Python 2.7 bridge architecture was created to overcome NAOqi incompatibilities. The system integrated OpenAI Whisper for speech recognition, GPT-4o for NLP, OpenCV for face detection and Flask for the Pepper tablet interface. The robot receptionist was tested on five participants who completed structured usability testing followed by an anonymous questionnaire. All functional and non-functional requirements were at least partially met, and all usability testing tasks were completed successfully. The system had high questionnaire scores for accuracy 4.4 / 5, trust 4.6 / 5, clarity 4.8/5 and 80% of participants preferred the robot over waiting for a human receptionist. The results demonstrate the viability of autonomous robot receptionists and show improvement over prior Pepper-based implementations through LLM integration. The key limitations were the fixed recording window, no conversational memory and a small sample of participants tested in a controlled laboratory environment. Future work could include true voice activity detection, conversational memory and larger scale testing.

Keywords: pepper robot, robot receptionist, social robotics, natural language processing, human-robot interaction

List of Acronyms

SMEs	Small or Medium Sized Enterprises
DOF	Degrees of Freedom
SLAM	Simultaneous Localization and Mapping
GUI	Graphical User Interface
LLMs	Large Language Models
ASP	Automated Social Presence
PIS	Participant Information Sheet
API	Application Programming Interface
SDK	Software development kit
FAQ	Frequently Asked Questions
NLP	Natural language processing
SSH	Secure Shell
JSON	JavaScript Object Notation

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1 Introduction

Automation is rising in all sectors of the UK as society becomes increasingly technology driven. This is affecting even service industries which have long relied on humans to run them and provide basic services. This has been a response to issues as more places are being expected to run 24 hours a day.

Currently, human receptionists face fatigue from working long unsociable hours. They are required to answer repetitive questions which could easily be answered without their input. They also face language barriers which can make it difficult to help people at the reception. Having to work long night shifts during unsociable hours can reduce their quality of life and happiness. It is also difficult on small businesses to hire 24/7 staff as they are constantly having to pay people and may have to pay premiums to attract less desirable night hours which can significantly reduce their profit margins.

A measure to address this could be an autonomous robot. It could be capable of greeting visitors emotively such as using pre-defined movement routines to mimic excitement to create happiness for visitors. It could maintain a high energy and create a good atmosphere that would be unrealistic for a human to be able to do continuously. It could provide different information to visitors such as the events taking place or what time the building closes which would reduce the amount of repetitive information and replies that human receptionists would otherwise have to provide. It could also more accurately give directions; by removing the human element of uncertainty, the robot would have a map of the rooms and know approximately where each room is resulting in less visitors being lost or arriving late.

1.1 Objectives

To address these challenges this project developed an autonomous robot receptionist using the Pepper robot. The objectives of this project were:

1. To design and develop an autonomous robot receptionist system using the Pepper platform
2. To integrate artificial intelligence tools to enable natural language interaction with visitors
3. To evaluate the system through structured usability testing with real participants
4. To assess the viability of social robots as a solution to challenges faced by human receptionists

2 Related work (Literature review)

The literature review begins by establishing the business case for why automation is needed and relevant in the service industry. Current robots struggle to fulfil the human element of interaction with visitors and anthropomorphic design explains why physical appearance helps to increase user acceptance transforming what could be an impersonal interaction into an engaging one. This project sought to move beyond the constrained systems visitors typically interact with such as a choice of a few buttons or a keyword system. LLMs like GPT-4o allow the flexibility to answer any questions allowing it to act as an intelligent assistant rather than just a simple tool (Wilske and Kruijff, 2006). Prior implementations have shown what has been tried, what has worked and what didn't, directly motivating the approach of this project. The deployment of such a system in a public environment requires careful consideration of handling data and deploying AI in a public space with real users and therefore an ethical framework was developed to ensure a smooth deployment.

2.1 The need for automation in service

The necessity for automation is driven by increasing demand for 24/7 availability. Providing 168 hours of weekly coverage requires multiple staff rotations creating a significant burden for small or medium sized enterprises (SMEs). Tuomi et al. (2021) argue that social robots could be used to change hospitality and reduce the amount of repetitive and mundane tasks using robots. While static automation such as McDonald's self-service kiosks or online chatbots have streamlined regular transactions they lack a social aspect needed for hospitality that could be filled by social robots. Van Doorn et al. (2016) developed a theoretical model comparing different service technologies by their social presence. The model found that social robots have a higher automated social presence (ASP) than kiosks due to their humanoid appearance and interactive behaviour. Social robots are viewed as a "social entity" by visitors as they can engage customers on a social level compared to kiosks which are viewed as utilitarian objects and this distinction supports the use of a robot over a kiosk for reception duties where social interaction is important. Replacing a human receptionist requires more than just functional software it requires a form that users are comfortable interacting with.

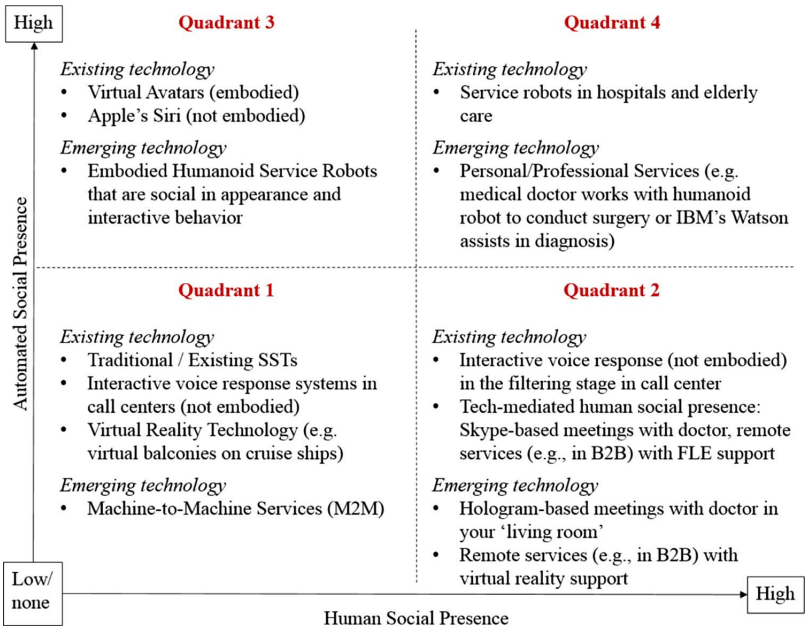


Figure 1 - Table showing how technology fits into ASP and human social presence from (Van Doorn et al. 2016)

2.2 Anthropomorphism and social design

A robot's physical behaviour is just as important as its code. A robot with movement and personality makes them appear more "approachable" and "trustworthy" than a regular machine giving it social presence. According to Xin and Liu (2025) "anthropomorphism in service robots by using human-like characteristics, motivations, intentions and emotions enhances human comprehension of robotic actions and promotes meaningful social interactions". This shows that user acceptance is higher when robots display familiar characteristics. It can communicate through non-verbal cues making it easier for users to understand and more engaging for them. From this research the project will incorporate emotive movement routines to mimic certain behaviours such as an excited greeting routine. By using anthropomorphic principles, the robot can increase user comfort and reduce the intimidation associated with autonomous systems. However, a physical presence alone isn't enough, the robot would need intelligence to engage in natural dialogue with visitors.

2.3 Natural language processing (NLP) and communication challenges

People don't always speak clearly, and the robot would need to map "Where is my coat?" to the same intent as "Where is the cloakroom?". If the user asks another question like "Where is that?" the robot needs to remember the context of what "that" refers to. The current industry solution is to map keywords to certain functions however large language models (LLMs) are replacing this to make the conversation more natural. Wilske and Kruijff (2006) argue this is because "people don't always mean what they say ... understanding the motives behind what someone says is crucial to successful interaction". Recognising the importance of interpreting what the visitor will say will mean the robot will not only be more successful with responding with useful verbal information but may have more flexibility to respond with appropriate non-verbal cues such as greetings or farewells. However, Wilske and Kruijff's work predates LLMs which have since represented a significant advancement in natural language understanding. Keyword mapping is now considered a limited approach compared to models such as GPT-4o which can interpret conversational intent without explicit rule mapping allowing for flexibility and is why it has been chosen for this project.

2.4 Learning from previous social robots in receptions

An example of a real-world implementation of robots in a reception setting is seen in the HOSPER study ran by Amabili et al. (2026). In this study, they used Pepper robots in a geriatric hospital to help provide directions to the elderly patients. The researchers found that Pepper displayed an elevated level of ASP through its anthropomorphic appearance and physical presence. It was adopted and easily interacted with by the elderly patients showing a reduction in "technological anxiety" that would usually be found in traditional kiosks. This is because it doesn't require any complex inputs and they can speak to the robots as if they were a human receptionist. However, the study noted a critical limitation was the robots lack of conversational flexibility. This is because of the robot's rigid pre-defined code meaning it was limited in responses to visitors and struggled to respond to complex requests. The lack of adaptability hindered their view of it as a "social entity". Consequently, this project aims to integrate an LLM to ensure the robot doesn't only give static replies but can fluidly give helpful information and humanlike responses. An issue with the HOSPER study was that it accepted conversational rigidity as a limitation rather than attempting to address it. This project directly targets that gap through LLM integration although whether it fully resolves the social entity perception issue identified by Amabili et al. (2026) remains to be seen and will be evaluated through user testing.

Beyond hospital settings, Pepper has been deployed outside a restaurant to study how it interacts with guests (Preusse et al., 2021). Initially, it acted as a host checking in guests through its tablet. However, an issue arose with it being unable to detect faces through face masks and was switched to a "greeter/entertainer". As Pepper couldn't recognise faces it was unable to respond to verbal interactions. This study shows the limitation with face detection reliability although it could be argued this issue is no longer relevant due to the rarity of face masks now compared to when the study was published in the covid era. Furthermore, the use of the tablet to check in guests has been shown to be viable in a real-world commercial setting showing its validity as an alternative input method. Preusse's response to the face detection failure was reactive as they changed the robot's role after the problem occurred. This project however will take a more proactive approach by designing the fallback from the start and will include a persistent welcome screen (Figure 23) irrespective of detection with check in and navigation buttons to maintain core functionality of the system without requiring a role change.

2.5 Ethical frameworks in service robots

The deployment of social robots in public spaces requires an extensive ethical framework. This is particularly concerning data governance, informed consent and the psychological impact of the anthropomorphic design.

2.5.1 Data privacy and digital paper trail

The difficulty with this project is finding a balance between needing a "smart" AI which requires sending data to the cloud and external companies such as OpenAI while maintaining the law in terms of GDPR and the Data Protection Act 2018 (UK Government, 2018). Under these laws, a person's voice isn't just a sound its potential biometric data that is personal to them and could be used for nefarious purposes and needs explicit consent. A person's voice data you can be used to clone their voice and impersonate them for bank security calls to gain access to their accounts or to manipulate their family members for gain. Therefore, the project will focus on data minimisation approach where only the minimal personal or identifiable information is collected, and data is anonymised to ensure legal compliance and trust with users. There is a risk from third party processing however the system processed speech to text and face detection locally meaning that the data never leaves the laptop/robot so that no biometric data leaves and it will be deleted immediately. This means external companies such as OpenAI will only receive text which would not be identifiable.

2.5.2 The right to know.

Ethical design interactions require that participants must know that they are being recorded and what they are agreeing to. They can receive this information through the Participant Information Sheet (PIS), consent form, and recruitment message. It is important that the robot is transparent when it is recording so that the participant does not disclose personal information. The participant information sheet makes sure the user is informed of what the data is being used for. Voluntary participation will be used so that users can be allowed to walk away at any moment for any reason to ensure that they are comfortable and are not distressed through participation.

2.5.3 The psychology of deception

While the robot's humanoid features and planned excited movements could make it feel more natural, it risks emotional deception. If the robot mimics human movements and thinking too closely it may trick the user into thinking it's more sentient than it is or that it could be used as a confidant with their personal information. Therefore, the project will counteract this risk with verbal reminders that it is a robot assistant to avoid misleading visitors. However, if the robot looks or acts too much like a person but fails to be perfect it can become creepy. This is what Mori, Macdorman and Kageki (2012) call the "Uncanny valley". Even though the Pepper robot was assigned to this project from the start it solves this problem as it looks like a stylised plastic robot rather than a fake human meaning it stays on the safe side of the valley. This means that it can avoid the "eeriness" that Mori warns about, allowing the user to enjoy the interaction while clearly understanding they are talking to a machine not a person.

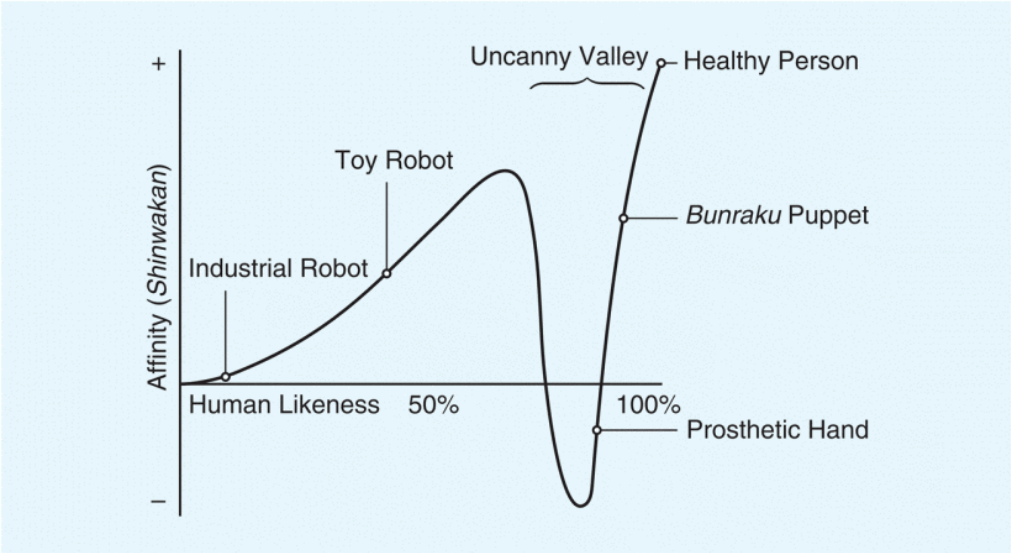


Figure 2 - Graph depicting the uncanny valley based on human likeness and human affinity with it based on certain objects.

2.5.4 Institutional oversight

This project was reviewed and gave favourable ethical opinion by the Ethics Committee at Aston University. The supervisor Ulysses Bernardet was responsible for overseeing this. Every participant in the study will be given a PIS sheet to ensure that they know what the study is for and how long it would take.

They will be given the option to ask questions about any concerns and will have a consent form to show they understood their data was being processed. The study was intentionally kept low-tech to ensure participant anonymity just consisting of an anonymous online questionnaire and a few observations of the robot. None of the answers were linked to names and referring to participants would be done by a number to ensure confidentiality. The testing will happen in a controlled space inside the university in the Robotics laboratory to ensure safety of participants and the robot.

2.6 Implications and implementation from the research

Having reviewed the relevant literature, several key findings directly informed the design decisions of this project. The literature review informed the project that social robots need more than functional software; they need an anthropomorphic design to be accepted by the visitors (Xin and Liu, 2025; Tuomi et al., 2021). It showed that LLMs are superior to keyword mapping for natural conversation (Wilske and Kruijff, 2006). Furthermore, previous systems such as HOSPER (Amabili et al., 2026) were limited by rigid pre-defined responses and LLM integration would solve this. GPT-4o was chosen over keyword mapping to address the conversational flexibility limitations. Ego-noise was found to be a real limitation of Pepper's microphone (Li et al., 2024) and the tablet GUI will be implemented as an alternative input method to mitigate this. SLAM navigation is noted to be unreliable by Mishra et al. (2023) on Pepper and therefore its movements will be limited and it will give verbal directions and show a map rather than physical navigation. The ethical framework is needed to ensure data privacy and informed consent and therefore the speech and face detection will be conducted locally to comply with GDPR and the Data Protection Act 2018 and minimise risk of data leakage. The uncanny valley must be considered in the design to ensure that visitors aren't intimidated or distressed by the robot (Mori et al., 2012) and Pepper avoids this issue through its stylised plastic design appearance that comfortably sits out of the uncanny valley.

3 System design and implementation

3.1 Methodology

This project followed the iterative development approach of Agile (Beck et al., 2001) rather than waterfall. Waterfall was unsuitable as the requirements were unknown upfront and changed frequently during development such as the Python 2.7 NAOqi incompatibility which wasn't discovered until implementation requiring the inclusion of the bridge architecture. This couldn't have been planned upfront so needed the flexibility to change to what the project required. Each feature was built and tested before moving onto the next one and influenced the next such as getting speech working before attempting to integrate NLP. Initially the Flask was created, then speech, then GPT, then face detection, then the bridge. Each feature was created as a new iteration adding a new feature that integrates with the others to create a working system. The original project definition form shown by Figure 24 in the appendix outlined the initial planned deliverables.

When the requirements changed the plan had to be adapted such as assuming NAOqi being able to automatically handle more functionality than it supported. Navigation was adapted to be more limited after SLAM unreliability was confirmed. Voice activity detection was simplified to a fixed-duration recording of 5 seconds due to microphone sensitivity constraints encountered during testing. The tablet scope was changed as it was originally intended to have more features however it was narrowed to just check-in and a map. Iterative suited this project specifically because hardware dependency meant testing was only possible on the physical robot. It wasn't possible to predict all the constraints in advance. As this was a solo project it meant there was no need for formal sprint ceremonies such as stand ups and retrospectives and instead this was based on a feature-by-feature basis.

3.1.1 Project planning

The project definition form (Figure 24) established the initial scope and timeline at the start of the project in October. This served as the initial ideas of what the project needed to contain and informed the project scope even where the timeline required changes. The functional and non-functional requirements in Section 3.2 served as the project backlog helping to prioritise what needed to be built. The key features were prioritised first such as the speech and face detection before more complex features like check-in. Progress was tracked against the planned timeline shown in Table 4 which is a Gantt chart detailing the planned and actual duration of tasks in the project and shows how the actual development changed from the plan, particularly the unplanned bridge architecture. The supervisor meetings documented in Table 5 served as informal checkpoints where progress was reviewed and direction adjusted. As this was a solo project there were no formal sprint ceremonies such as daily stand-ups and the progress was self-directed on a feature-by-feature basis with each completed feature informally tested before moving on.

3.2 Requirements

The use case diagram (Figure 3) illustrates the core interactions between the visitor, staff member and the system. This shows the core functionality that is needed for the system.

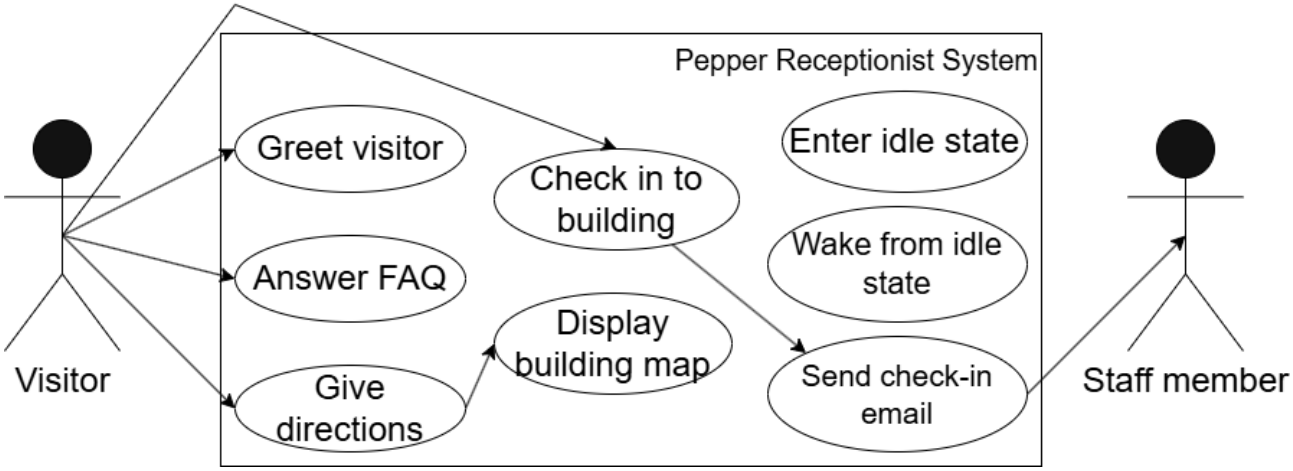


Figure 3 - Use case diagram showing interactions between system actors and the Pepper receptionist system

3.2.1 Functional requirements

F1. The system must detect a visitor using face detection.

F2. The robot must greet visitors with movement and speech.

F3. The system must record visitor speech via a microphone.

F4. The system must transcribe speech to text using Whisper.

F5. The system must process visitor intent using GPT-4o.

F6. The robot must respond verbally with a dynamically generated reply.

F7. The robot must display a check in form on tablet when requested.

F8. The system must send email notification to staff on check-in submission.

F9. The robot must display a building map on tablet for navigation queries.

F10. The system must manage multi-turn conversations up to five exchanges.

F11. The robot must enter an idle state after inactivity.

F12. The robot must wake automatically when visitor detected.

3.2.2 Non-functional requirements

NF1. The system must comply with GDPR (No biometric data leaving laptop)
NF2. The system must respond in a reasonable time to maintain natural conversation flow.
NF3. The system must handle one visitor interaction at a time.
NF4. The system must remain stable through a session without requiring a manual restart.
NF5. The system must be accessible to users without prior experience of social robots.
NF6. The robot's response must be appropriate and responsible for a university reception environment.

3.3 Technical capabilities and limiting factors.

The Pepper robot has many technical capabilities and limiting factors and it is important to consider how these affect its design. Pepper is "currently one of the most widely adopted service robots in the world" according to Tuomi et al. (2021), and it has proven results in many businesses and in terms of research. The key components listed on Maxtronics are perception sensors including cameras to view its surroundings and microphones to listen to visitors. It has actuators that give it degrees of freedom (DOF) allowing for fluid movements from multiple interconnected joints. It also has omnidirectional mobile wheels allowing for it to quickly turn to face visitors. In a technical study conducted by Mishra et al. (2023) it was found that its "sound and speech capabilities yield satisfactory results," however its "built-in Simultaneous Localization and Mapping (SLAM) navigation is unreliable making it difficult to reach destinations accurately due to generated maps". Taking these limiting factors into consideration the robot's navigation was therefore limited to stationary anthropomorphic movements. This also better mimics a human receptionist where a visitor walks over to ask a question rather than being approached which could be intimidating or unnecessary. Multiple voices or sounds could also be an issue. According to Li et al. (2024) "social robots currently cannot prioritize particular sounds. More specifically, they generally lack the ability to extract and recognize human speech when they are talking themselves... miscommunication occurs when a user starts answering before the robot finishes, as parts of the answers will be left out." This is called ego-noise and is one example of how voice input from visitors could be cut off. To mitigate ego-noise, the tablet graphical user interface (GUI) provides an alternative input method ensuring visitors can still interact if voice recognition fails. One major limiting factor is that Pepper's manufacturer Aldebaran Robotics went into receivership raising questions about its future. This means that it may not have long-term viability however the development of this system could be applied to many different systems.

3.4 Software architecture and tools

Table 1 - Key tools and their purpose in the project

Function	Tool	Purpose in the project
Coding language	Python	Primary language to control the robot
Speech to text	Open AI Whisper	Convert speech into text to allow the robot to understand
Natural language processing	GPT-4o	Used to understand visitor speech and their intent
Text to speech	ALTextToSpeech	Text to speech library to give Pepper a voice
Interface	Flask	To create interface on Pepper's tablet
Vision	OpenCV	Face-detection to look at guests

A core architectural challenge was that NAOqi, Pepper's application programming interface (API) is restricted to Python 2.7 while modern AI libraries such as Whisper and the OpenAI software development kit (SDK) require Python 3. Rather than downgrading the AI stack or porting NAOqi, a bridge architecture was designed to run the two environments as separate processes. A TCP socket on port 6000 was chosen as a communication channel as it is lightweight requiring no additional libraries and works reliably across separate processes on the same machine. JavaScript Object Notation (JSON) was selected as the message format as it is natively supported in both Python versions and is easily readable simplifying debugging. The bridge was kept as a separate process rather than a thread because NAOqi blocks the execution during robot commands stalling the AI pipeline if they shared a process.

The system is launched via a batch script which starts four concurrent processes in sequence: the Flask web server, a secure shell (SSH) startup script on the Pepper itself, the Python 2.7 bridge, and the main Python 3 loop. A two second delay between each process ensures that each script's dependencies are ready before the next component begins. Originally the terminals were run manually by opening the terminals and pasting the commands, but this was changed to include the batch script to increase speed and consistency with testing. The main loop was also updated to wait for the bridge to be ready before starting. This fixed a timing problem from earlier testing where processes initialised before their dependencies were ready resulting in the greeting not occurring. This slowed down Pepper's startup time but greatly improved its consistency.

The Flask web server runs on the laptop and can be accessed on the local network. It serves the visitor check-in form alongside its logs as well as the main building map on Pepper's tablet. Rather than running Flask on Pepper directly, it was hosted on the laptop avoiding constraints and complications involved with its internal hardware and software and to allow easier debugging and modifications.

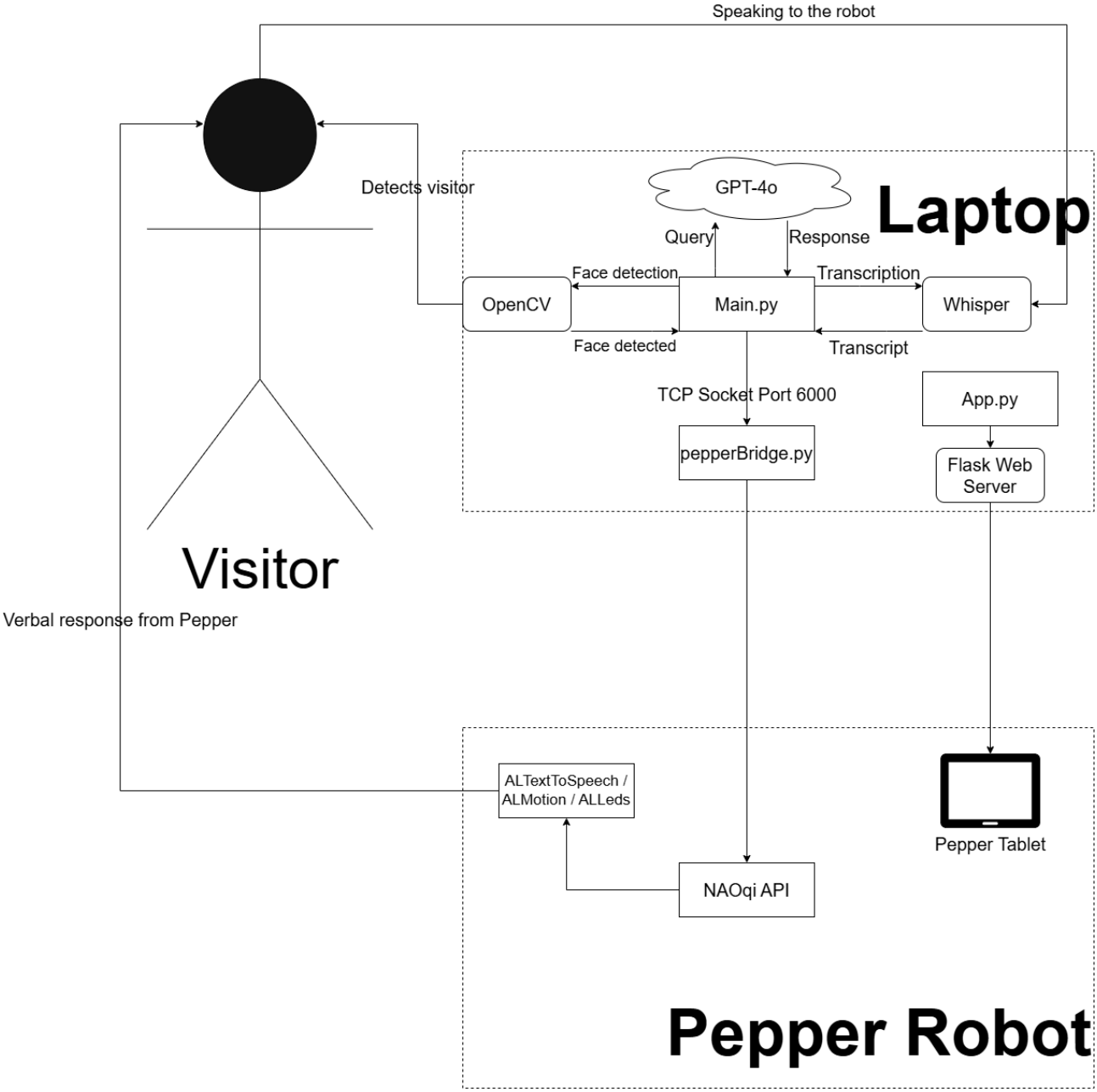


Figure 4 - System architecture diagram showing component interactions

3.5 System interaction flow

There is a five-stage pipeline that controls each visitor interaction. In the detection stage, OpenCV continually processes the camera feed using a Haar cascade classifier to identify faces within range. Once a visitor is detected, Pepper initiates a greeting stage involving arm raises, flashing eye LEDs and a wave before delivering a welcome message via ALTextToSpeech. The system then enters the listening stage where audio is recorded by laptop microphone and passed to the Whisper medium model. Whisper transcribes the audio to text which is passed onto the thinking stage. The thinking stage starts and the text is sent to GPT-4o. GPT-4o returns JSON with two fields: the intent and the reply. The intent determines what happens next in terms of greeting, checking in, frequently asked questions (FAQ), navigation, unknown, and goodbye. Pepper then delivers the reply via ALTextToSpeech for the responding stage which is dynamically generated by the AI to be relevant. This information is determined by a large amount of specialist knowledge stored in the FAQ section of the prompt giving Aston University specific information such as how to find certain rooms. The tablet displays relevant content when applicable such as the check-in form or the main building map. The system loops between the listening and thinking stage up to five times to answer any questions. The conversation ends automatically after five exchanges or if the visitor says goodbye or similar. The sequence flow diagram (Figure 5) shows the order of messages between components in the system during a typical visitor interaction. The flowchart (Figure 6) provides a visual overview of the full system flow and shows decision points and branching logic.

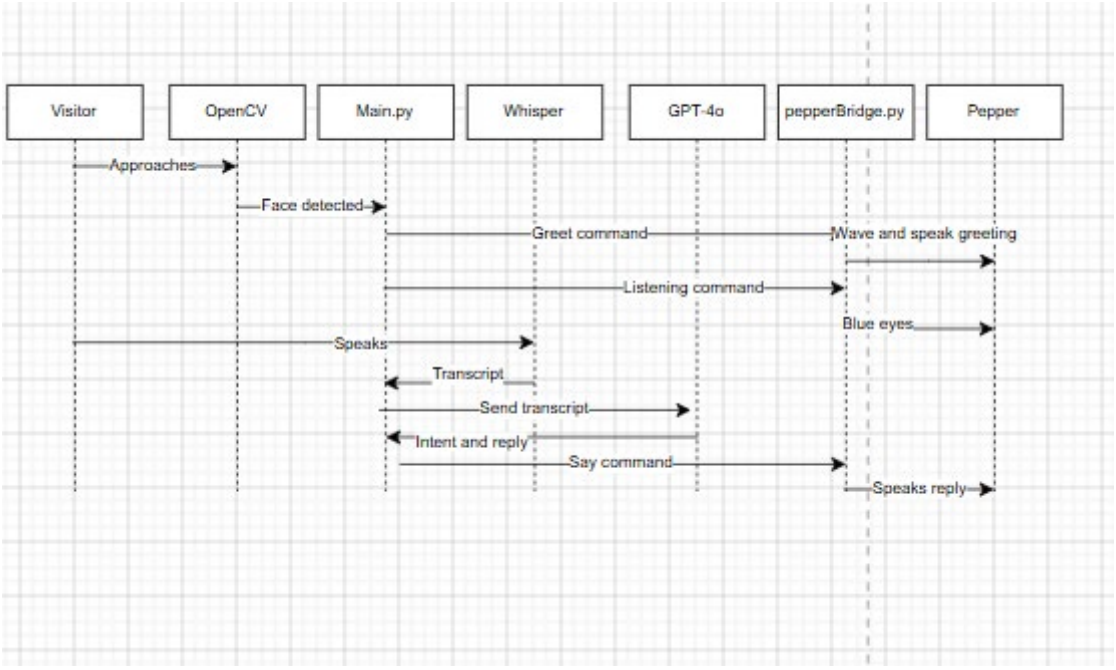


Figure 5 - Sequence flow diagram showing message flow during a visitor interaction

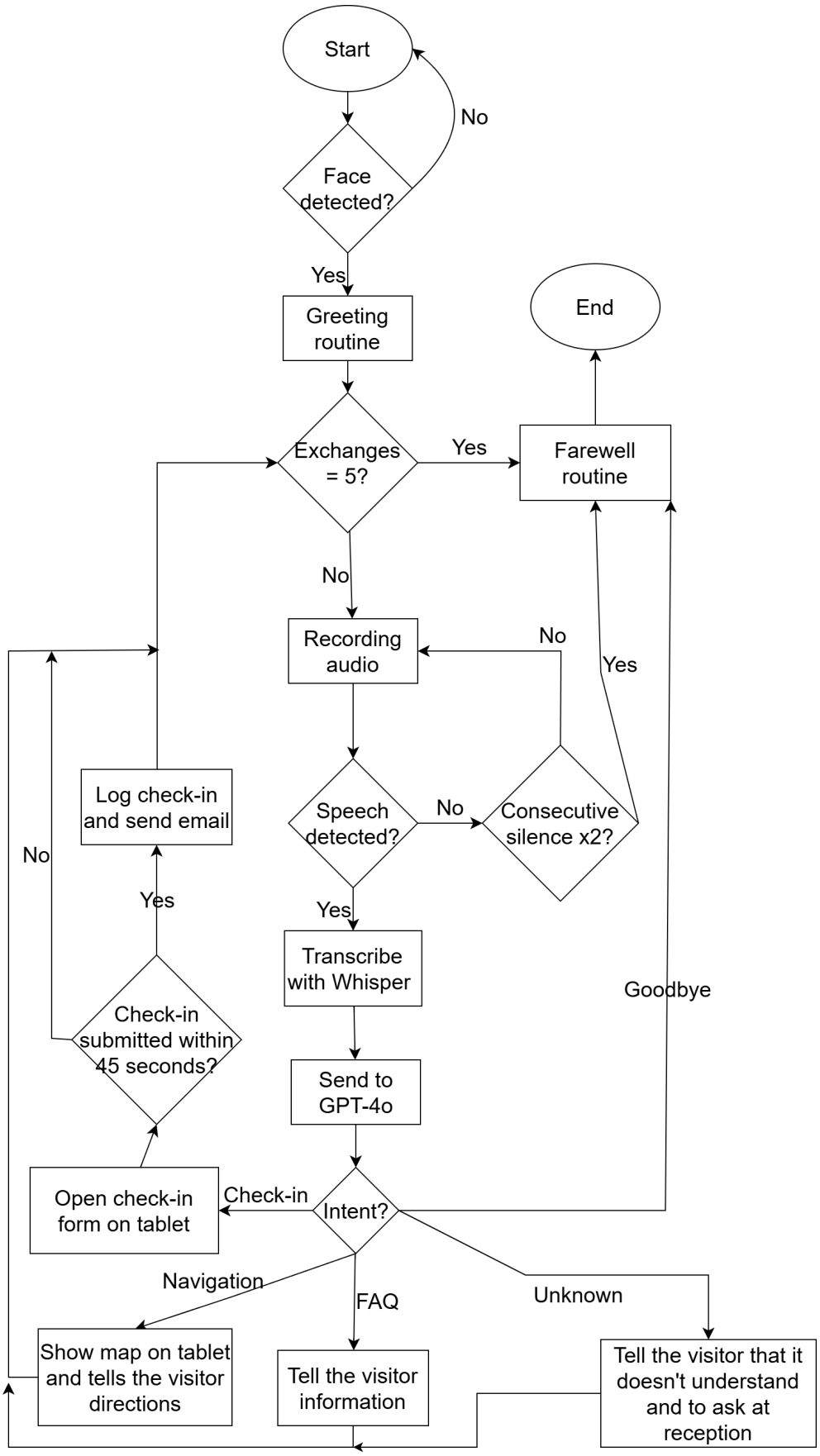


Figure 6 - Flowchart showing the system interaction flow

3.6 Conversation loop and robot states

The Pepper robot is programmed with six distinct states each with a matching eye LED colour to clearly display what the robot is doing. For example, white shows its standby state, blue shows its listening state, yellow its thinking state and green its responding state. Colour coding gives visitors non-verbal feedback on whether the robot has heard them. This helps to directly address the ego-noise limitation mentioned in Section 3.3 as visual feedback helps to compensate when audio is uncertain. There are further links to anthropomorphism literature discussed in Section 2.2 that non-verbal cues help make interaction feel more natural and mimics how human facial expressions give non-verbal clues about their intent (Xin and Liu, 2025).

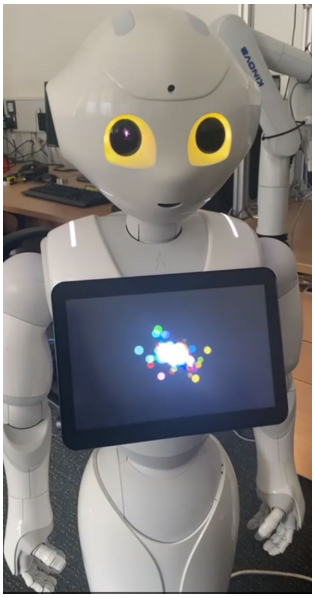


Figure 7 - Pepper showing the thinking state through its yellow eyes

A sleep and wake cycle has been integrated into the system where if no face has been detected for thirty seconds, then it triggers a sleep state. In the sleep state its head droops, arms lower and eyes dim to blue to show that it is idle. Upon waking, its eyes flash and it returns to a standing posture automatically when a visitor's face is detected. This mimics a human receptionist being visibly idle versus actively engaged. This helps to save energy and signals availability to passing visitors.

To help manage conversations the system has been limited to five exchanges before an automatic farewell. If two consecutive exchanges are silent, then it assumes the visitor has left to prevent it from constantly asking questions. Transcript cleaning has been implemented to help prevent the system mishearing certain phrases. In particular, the system struggled with short alphanumeric codes for Aston University room numbers, so it used replacements such as “em be” equating to “MB” to ensure that visitors are not misheard. A hallucination filter was also implemented to discard known false-positive transcriptions from Whisper, such as the phrase “Thank you” being generated when there was no speech detected ensuring that these did not reach GPT-4o and cause incorrect responses. A 45 second timeout on the check-in form was implemented on the assumption that the visitor has abandoned it. File-based flags in sharedState.py were used to signal between Flask and the main loop without blocking either process. When the visitor submits or cancels a form, Flask writes a flag file which the main loop repeatedly checks until it finds it. Originally this was done through threading however due Flask and the main loop being run as separate operating system (OS) processes they couldn’t share memory.



Figure 8 - Evidence of common hallucinations being handled by the system

To process faces OpenCV is used to process every frame of the camera feed. A Boolean flag (visitor_present) prevents the interaction triggering repeatedly on consecutive frames for the same visitor. The interaction triggers once per visit not once per frame to reduce errors from overlapping processes. When there is no face detected the Boolean flag resets so that it’s ready for the next visitor. The detection window displays on the operator’s laptop showing a green box and “Visitor Detected” label. This was useful during development and testing to verify that the detection was working correctly on visitor faces.

3.7 Check-in system

Checking in is a major component of many receptionist's duties. The check-in system works through the visitor speaking to the robot and GPT-4o returning a check-in intent based on what they said. Pepper directed visitors to the tablet and opened a Flask web interface with the check-in form. The form collects the name, student/staff ID, the staff member being visited and the purpose. On submission the visitors' details are logged onto another Flask webpage, and an email is sent to the relevant staff member. The email is sent on a background thread so that the conversation loop is not blocked and the visitor can keep interacting with the robot. A file-based flag event in sharedState.py signals when the form is submitted or cancelled to close the check-in form. It has a clear button to reset the form in case the visitor has mistakes or changes their mind. There is a forty-five second timeout before the system assumes the visitor has abandoned the form and cancels it. The visitor log is accessible via the /log route on the Flask server to see a table of who is visiting. The staff directory is hardcoded in app.py so the visitor can just select from a dropdown menu to send an email to a specific staff member increasing simplicity. During testing however email notifications were routed to a personal email address due to university emails experiencing significant delays due to quarantine filtering on unrecognised senders (Figure 10). This component automates a repetitive receptionist task (Tuomi et al., 2021) and helps to notify staff in real time without human involvement to instantly inform them of visitors by email.



Figure 9 - Check-in screen on the Pepper robot

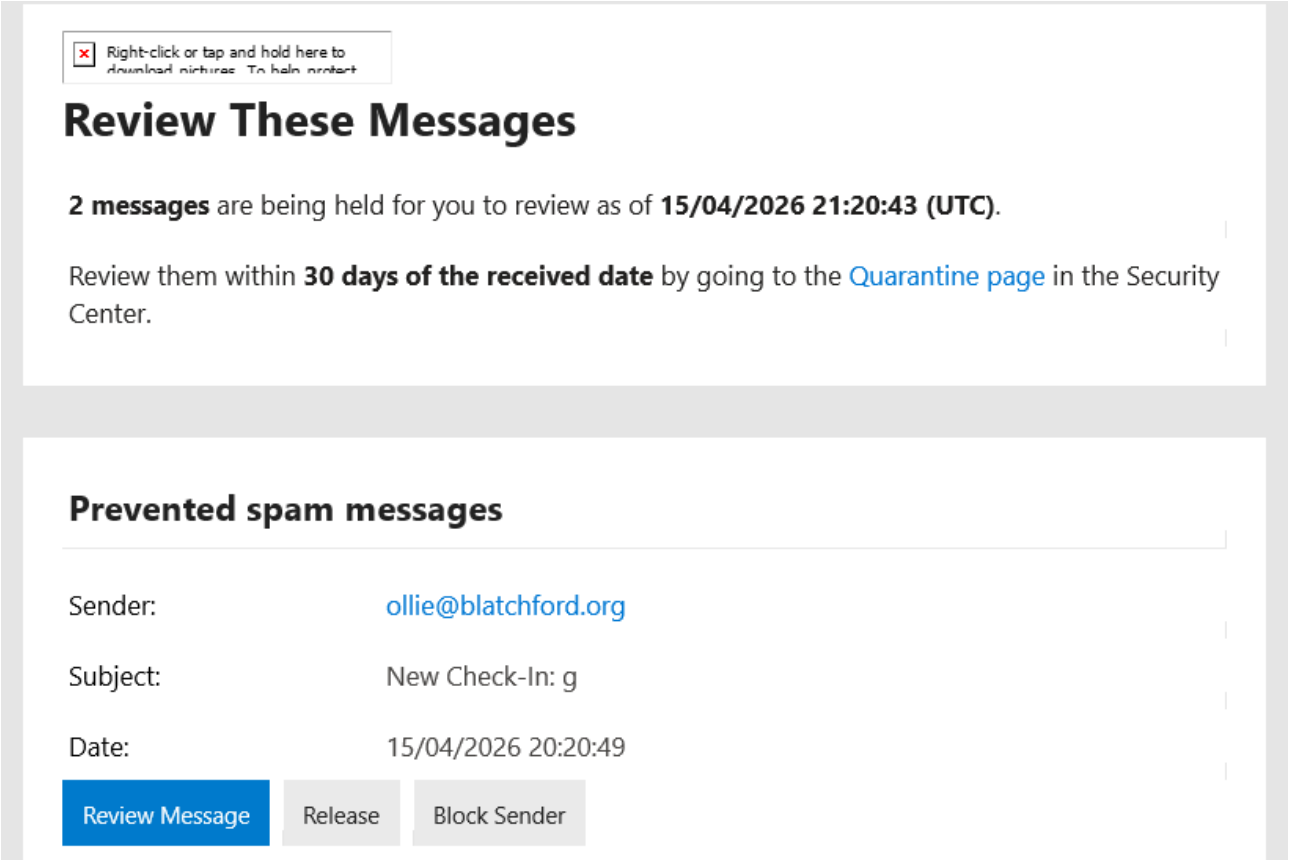


Figure 10 - Quarantined email evidence

4 Evaluation

The system was evaluated through two methods. Firstly, a structured usability test (Figure 25) where participants completed five usability tasks interacting with Pepper in the robotics lab such as asking for directions. Secondly, an anonymous questionnaire gathering feedback on the interaction with Pepper. Each task is discussed in turn followed by a questionnaire analysis. The requirements detailed in Section 3.2 are evaluated based on the evidence gathered to assess the success of the system.

4.1 System testing overview

The system was tested in the Aston University robotics laboratory (MB306). Pepper connected via university Wi-Fi, shared by a mobile hotspot from the laptop (Figure 12). The system ran across four concurrent terminals: the Flask web server, the NAO startup, the Python 2.7 bridge and the main Python 3 loop (Figure 11). The laptop ran Windows with the Python 3 virtual environment and Python 2.7 installed separately.

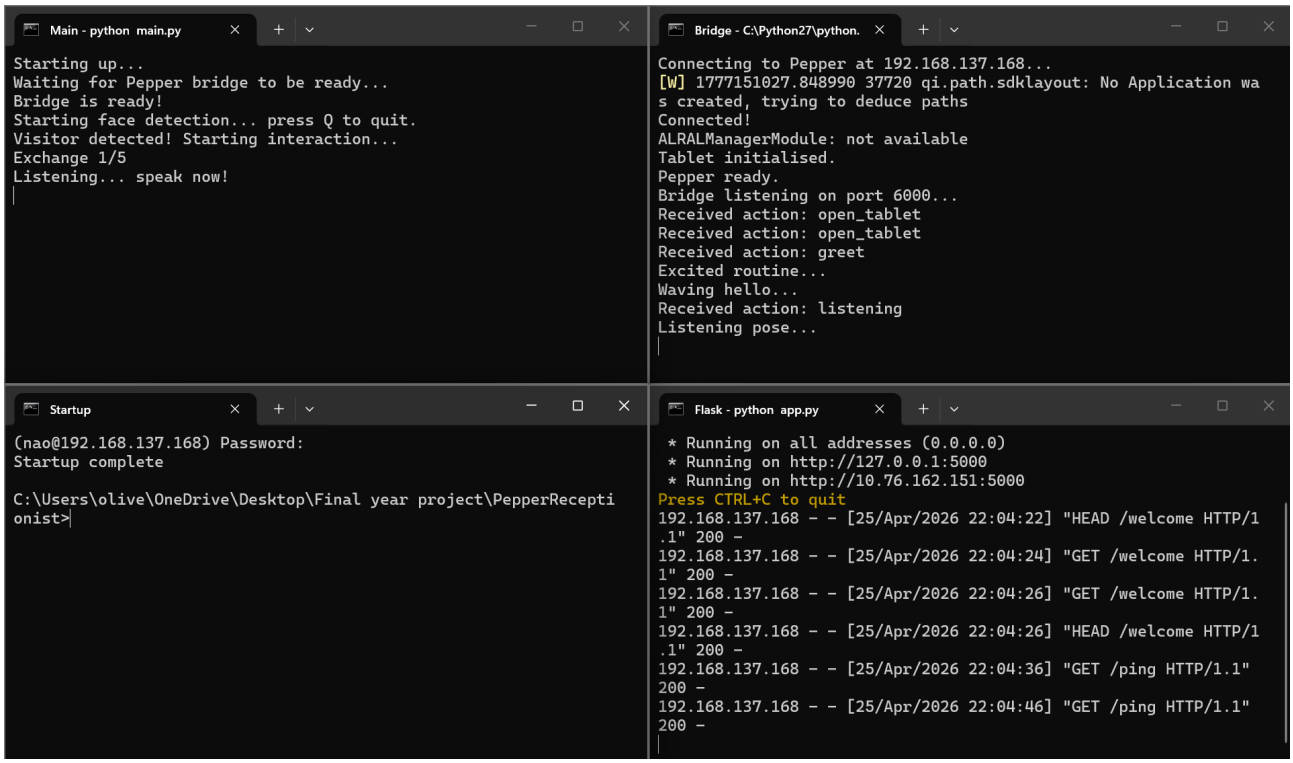


Figure 11 - The four terminals running the system simultaneously



Figure 12 - Mobile hotspot showing Pepper connected to laptop

Five participants took part in the usability testing. Participants approached the robot naturally as they would a real receptionist. Each participant was sent a PIS sheet and consent form via email before the testing session. Participants were asked to interact with the robot following the usability test task document. The logs of the interaction in the terminal were recorded. They were given an anonymous questionnaire link in Google Forms to complete at their convenience. It never crashed during the testing however it did sometimes have small malfunctions such as not saying the greeting message or the tablet not connecting properly. Sometimes participants also missed when they were meant to speak resulting in it asking them to ask again. It did not require restarts between participants. The participants showed a mix of emotions either excited to be in the presence of a robot or a little unnerved by its humanoid nature and its staring. The excited greeting landed well.



Figure 13 - Picture of Pepper setup in the laboratory plugged into its charger

4.2 Task 1 – Saying hello

The purpose of this task was to test whether face detection reliably triggers the greeting and whether the anthropomorphic greeting routine makes visitors feel welcomed. Face detection triggered correctly when participants approached Pepper as seen by Figure 14. Pepper began its excited greeting routine by raising its arms, flashing its LED eyes and waving, introducing itself and welcoming the visitor to Aston University. All five participants triggered the greeting successfully and were greeted by Pepper and appeared to be engaged by it, consistent with the anthropomorphic design principles discussed in Section 2.2.

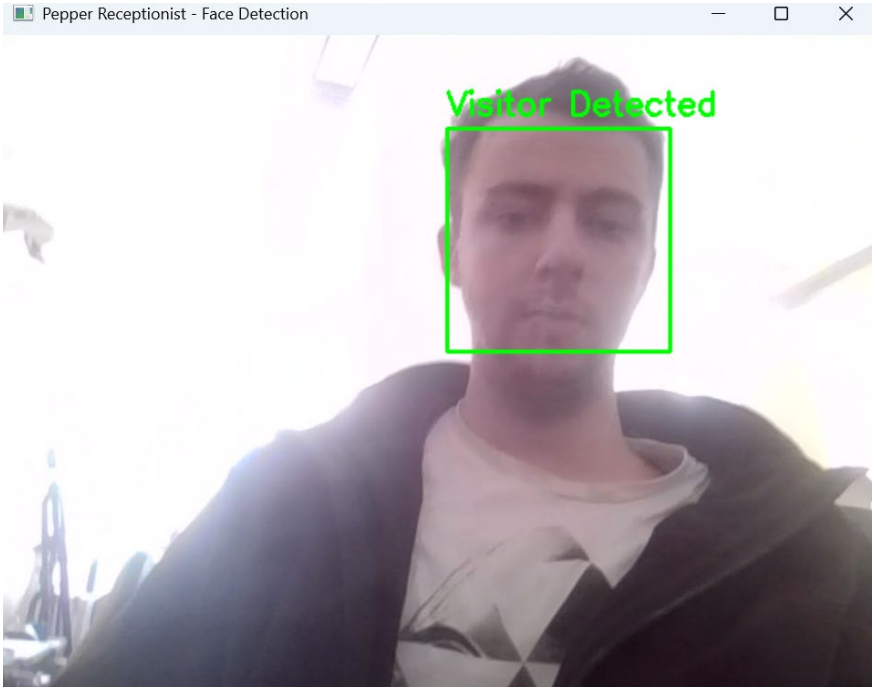


Figure 14 - Face detection by the system highlighting the visitors face with a green box saying visitor detected



Figure 15 - Greeting routine upon seeing a visitor's face

4.3 Task 2- Finding your way

Task 2 tested whether GPT-4o would correctly return the navigation intent and the accuracy and helpfulness of its verbal directions (Figure 16) and non-verbal directions (Figure 17). Participants asked for directions to an upper floor from Pepper and how to get there such as the location of the stairs or name of the lift. GPT-4o correctly identified the navigation intent and returned it. As seen in Figure 16, Pepper gave verbal directions to get to the room such as room MB705 being on the 7th floor at the front left side of the building telling the user to use either the main stairs or the red/blue sky lifts near the reception. While giving the instructions a map of the ground floor of the main building showed on its tablet (Figure 17) to help the visitor locate the rooms and it lingered for around 10 seconds to give them time to look. The ability to give directions was a limitation in the HOSPER study in Section 2.4 due to rigid pre-defined responses however the LLM integration from this project addressed this.

```
Listening... speak now!
Recording saved.
Transcribing...
Visitor said: mb705.
Intent: navigation | Reply: Room MB705 is on the 7th floor, front left side. You can use the main staircase or the Red/Blue Sky Lifts
 near the reception.
```

Figure 16 - Main terminal showing correct Pepper response to directions



Figure 17 - Pepper display the main building map on its tablet

4.4 Task 3 – Getting information

This task involved testing the FAQ knowledge base in the system prompt and its sufficiency to answer common building questions accurately. Participants asked a variety of questions such as main building opening hours, the location of the reception and information about the library. The FAQ knowledge base contained the answers to most questions asked and they were accurate. However, one participant asked where they could get a pencil in the building which wasn't stored in the knowledge base. As a result, Pepper instructed them to speak to the main reception demonstrating that the system correctly defers visitors to human staff rather than fabricating responses when knowledge is unavailable. This task directly shows the capabilities of Pepper and the system to replace the repetitive question answering mentioned in Section 1.

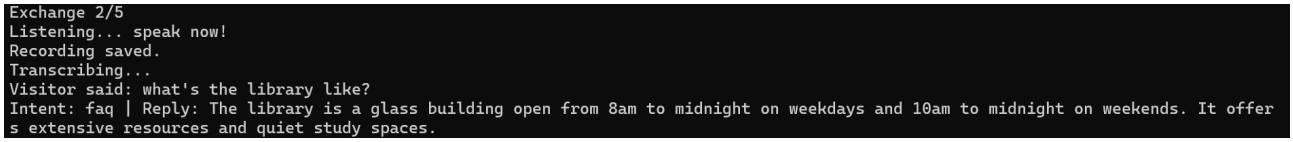


Figure 18 - Main terminal showing Pepper's response to a question asking information about the university

4.5 Task 4 – A harder question

Task 4 tested whether GPT-4o could handle the more complex open-ended questions being asked by visitors that require interpreting rather than direct keyword matching. Participants asked a variety of questions such as asking for a recommendation of places to study and places to get food. Pepper recommended a variety of study spaces such as the zen zone and directed participants to the Aston Coffee Lab for food. However, the system did not perform true reasoning beyond its knowledge base rather GPT-4o's natural language understanding allowed it to infer the meaning of entries in the FAQ such as the Coffee Lab being related to food. This helps solve the issue addressed in the HOSPER study in section 2.4 where Pepper struggled to handle complex requests but the integration of LLM allows it handle open-ended questions better, supporting Wilske and Kruijff's (2006) argument that understanding visitor intent is crucial for effective interaction.

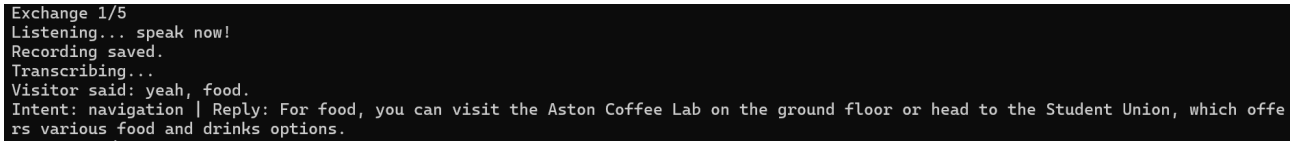


Figure 19 - Main terminal showing Pepper's response to being asked a harder question such as inferring that the Aston Coffee Lab serves food without the system prompt having information on this

4.6 Task 5 – Checking into the building

This task tested whether participants could use Pepper to check-in to the university and if it returned the check-in intent appropriately. The check-in takes approximately a minute to complete, and the participants found the form intuitive to a certain extent. The forty-five second timeout didn't cause any issues as the participants comfortably completed the form in the window. When participants asked to check in, GPT-4o returned the correct intent and directed visitors to use the tablet and a check-in form appeared (Figure 20). Participants filled in name, ID, staff member and purpose and the form submitted successfully. The visitor log (Figure 21) was updated with every entry and an email notification was sent to a personal Gmail address (Figure 22) due to complications with university email quarantining (Figure 10) which would be a real-world deployment consideration. All five participants successfully managed to check in through Pepper. This task directly demonstrates the automation of a core reception duty as argued in Section 1, reducing the burden of repetitive check-in tasks on human staff. Furthermore, the addition of a persistent welcome screen with buttons for check-in and map as an alternative input directly addresses the issue found in Preusse et al. (2021) where Pepper was required to change its role for checking in due to issues with face detection.

Two usability issues were identified during testing. Pepper occasionally moved its head to look at the visitor while they were typing causing the tablet to shift slightly making input more difficult. Another complication was that the on-screen keyboard sometimes obscured some of the form fields meaning participants took a moment to navigate around this. Despite these issues all five participants successfully completed the check-in process demonstrating that F7 and F8 were met.

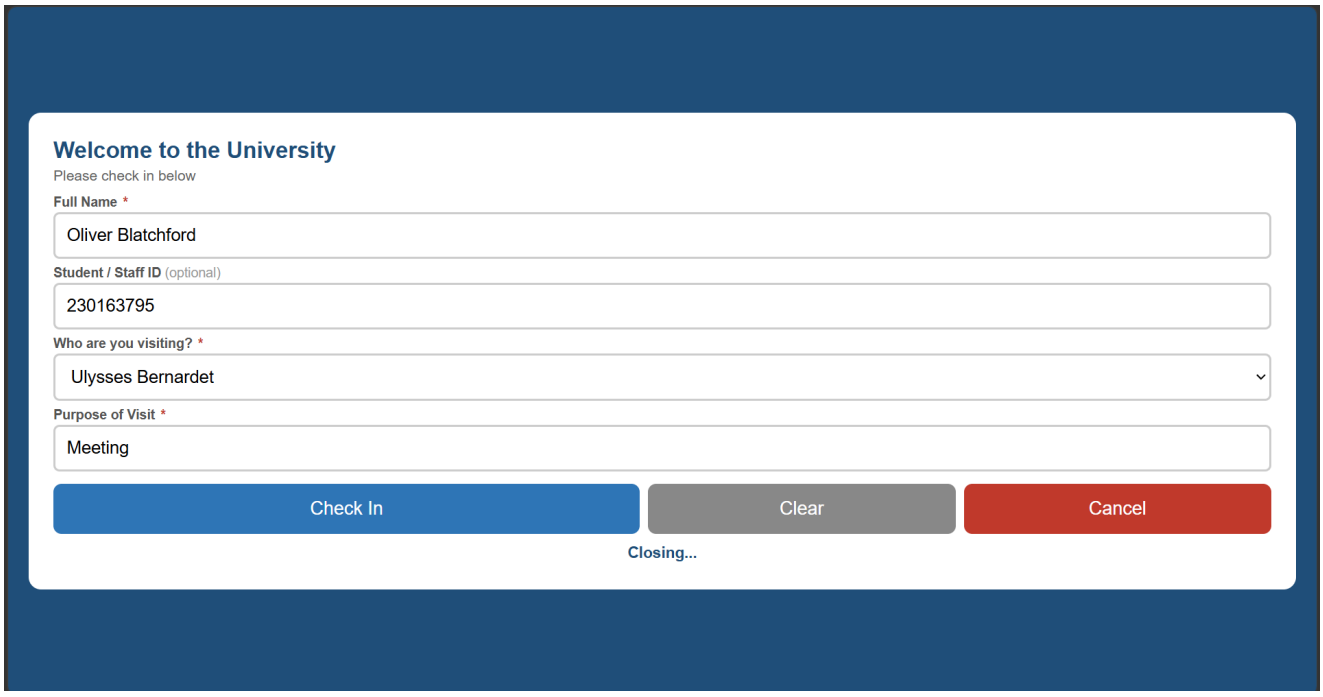


Figure 20 - Check-in form on Pepper tablet

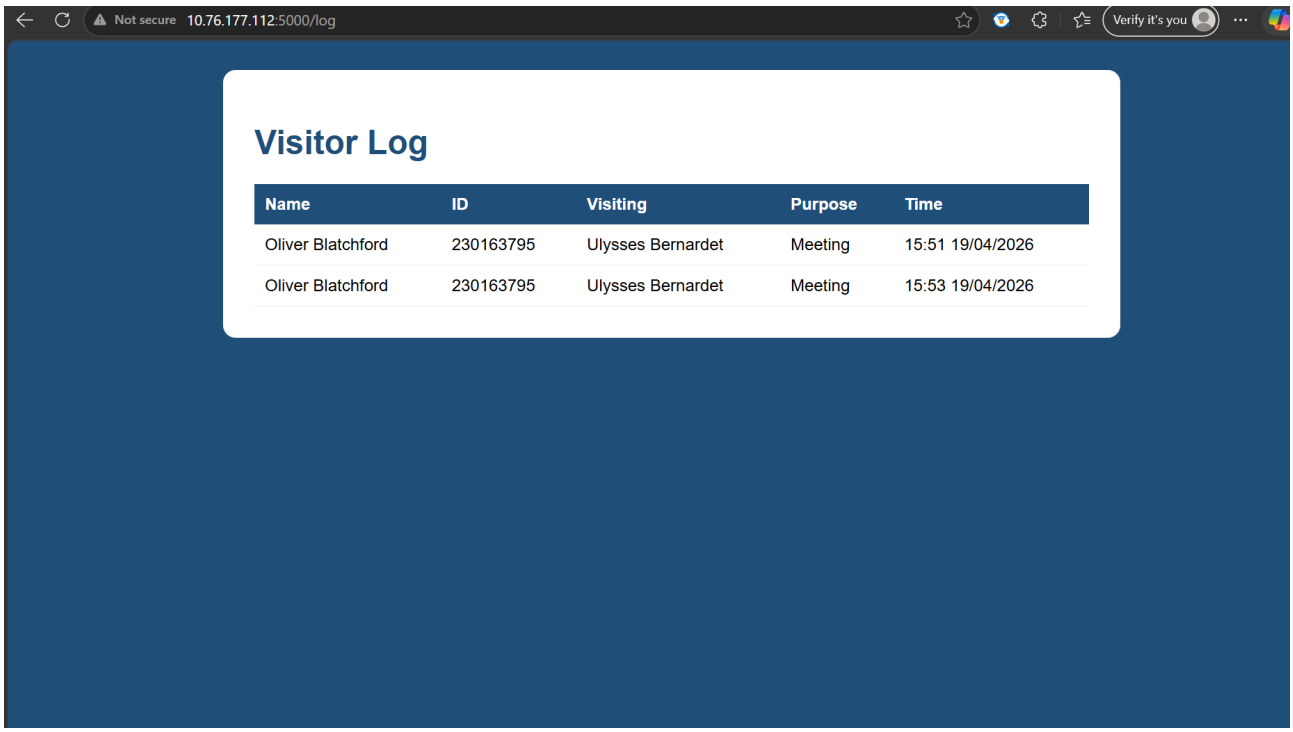


Figure 21 - Visitor log showing visitors that have checked in.

Oliver Blatchford is here to meet Ulysses Bernardet for Meeting.

Time: 15:53 19/04/2026
Student/Staff ID: 230163795

Figure 22 - Email sent by the system after a visitor has checked in

4.7 Questionnaire results

After interacting with Pepper, the participants completed an anonymous questionnaire on the experience covering response quality, comfort, system performance and suggestions for improvement. This section contains quantitative feedback through ranking numbers and qualitative feedback through written feedback on the participant experience. The questionnaire results can be found in Appendix 7.3.

The questionnaire highlighted that Pepper had good response quality with average scores for the following traits: accuracy 4.4 / 5 (Figure 27), trust 4.6 / 5 (Figure 29), clarity 4.8 / 5 (Figure 31) with no participant rating lower than a 3 for these. This suggests the FAQ knowledge base and GPT-4o produced accurate, trustworthy and clear responses. The high trust score is particularly notable as it means that participants believed what Pepper told them. High response quality demonstrates NF6 being fulfilled as participants found the responses from Pepper appropriate and responsible. These scores represent a quantitative improvement over prior implementations of Pepper such as HOSPER (Amabili et al., 2026) which was critically limited by rigid pre-defined responses and could not handle complex visitor requests.

User experience however showed more mixed results with movements 3.6 / 5 (Figure 28) and comfort 3.8 / 5 (Figure 30). The movements helped but no one rated the movements a 5 suggesting the routine was appreciated but could be more expressive and engaging. One participant rated the comfort as 2 suggesting some discomfort consistent with the uncanny valley effect mentioned by Mori et al. (2012) and is consistent with the observation in Section 4.1 that some participants were unnerved by the robot's staring meaning that NF5 is only partially met due to accessibility of the robot.

The system performance showed good results with 4 out of 5 participants reported Pepper understanding their request the first time with just one participant requiring 2 attempts (Figure 32). However, the response time had divided opinions with 60% finding it acceptable, 20% feeling neutral and 20% highlighting it as an issue (Figure 33). This shows NF2 being partially met as most participants found it acceptable but not all of them.

The overall preference of participants showed that 60% agreed they would prefer the robot, 20% were neutral and 20% strongly agreed (Figure 34). Nobody disagreed with this question validating the core argument from Section 1 about robots being able to reduce the burden on human receptionists and suggests that participants saw value in the system over waiting for human staff.

The participants provided qualitative feedback in Figure 35 and Figure 36. A key issue identified was the keyboard covering the boxes mentioned by two participants, validating usability issue identified in Task 5. Issues with the response speeds and the timing of when to speak were highlighted. Participants suggested features such as conversational memory (F10), faster responses (NF2), date/time display and more interactivity such as being able to shake Pepper's hand.

4.8 Evaluation against requirements

4.8.1 Functional requirements

Table 2 – Functional requirements of the system detailing their status and evidence

Requirement	Description	Status	Evidence
F1	The system must detect a visitor using face detection.	Met	Figure 14
F2	The robot must greet visitors with movement and speech.	Partially met – occasionally doesn't do the greeting if not started up correctly	Figure 15
F3	The system must record visitor speech via a microphone.	Met	Section 4.1

F4	The system must transcribe speech to text using Whisper.	Partially met – occasional mishearing and hallucinations handled by transcript cleaning	Figure 16
F5	The system must process visitor intent using GPT-4o.	Met	Figure 16
F6	The robot must respond verbally with a dynamically generated reply.	Met	Figure 16, Figure 18
F7	The robot must display a check in form on tablet when requested.	Partially met – tablet occasionally failed to connect if not started up correctly	Figure 20
F8	The system must send email notification to staff on check-in submission.	Partially met – email worked but university quarantine prevented delivery to staff	Figure 10,Figure 22
F9	The robot must display a building map on tablet for navigation queries.	Met	Figure 17
F10	The system must manage multi-turn conversations up to five exchanges.	Met – although no conversational memory	Figure 18,Figure 19
F11	The robot must enter an idle state after inactivity.	Partially met – occasionally doesn't trigger idle state	Section 3.6
F12	The robot must wake automatically when visitor detected.	Met	Figure 14, Figure 15

4.8.2 Non-functional requirements

Table 3 - Non-functional requirements of the system detailing their status and evidence

Requirement	Description	Status	Evidence
NF1	The system must comply with GDPR (No biometric data leaving laptop)	Met	Section 5.1
NF2	The system must respond in a reasonable time to maintain natural conversation flow.	Partially met - GPT-4o processing caused delays in conversation	Figure 33
NF3	The system must handle one visitor interaction at a time.	Met	Section 4.1
NF4	The system must remain stable through a session without requiring a manual restart.	Met – no crashes, minor malfunctions only	Section 4.1
NF5	The system must be accessible to users without prior experience of social robots.	Partially met – some users struggled with the timing of speaking for the robot, and some were unnerved by its staring	Section 4.7
NF6	The robot's response must be appropriate and responsible for a university reception environment.	Met	Section 4.7

5 Conclusion

The project successfully developed a working and autonomous robot receptionist system that demonstrated the viability of reducing the burden from human receptionists. The system integrates face detection, speech recognition, LLM-based NLP and a check-in system. All twelve functional and six non-functional requirements were at least partially met, with the majority fully met. The original project definition form (Figure 24) outlined several features that were descoped during development. Navigation and path finding were descoped due to SLAM unreliability; calendar integration was not implemented due to time constraints during development. The core receptionist functionality has been successfully delivered with a greeting, FAQ, directions and check-in. The iterative development approach from Agile allowed the project to adapt to its changes resulting in a functional system. Participant testing demonstrated that the system could reliably handle all five usability tasks with positive feedback on response accuracy, trust and clarity. The four objectives outlined in Section 1.1 were all addressed. The system was designed and developed using Pepper integrating AI tools such as Whisper and GPT-4o to enable natural language interaction. Structured usability testing with five participants demonstrated that all the tasks were completed successfully and the results validate the viability of robots as a solution to the challenges faced by human receptionists.

5.1 Strengths

The resulting system created from this project has many strengths. Pepper did not crash during the entire usability testing showing that the code is stable. The use of an LLM in GPT-4o has proved useful as it gave accurate, trusted and clear responses evidenced by high questionnaire scores in Section 4.7. The bridge architecture developed for this project helped solve a major issue of incompatibility and was a novel solution by enabling the use of modern AI libraries. The check-in system correctly submits a form, logs it and sends an email notification and was a good feature directly reducing receptionist responsibilities as argued in Section 1. The system addressed challenges from its predecessors such as HOSPER's conversational rigidity by implementing an LLM to allow it to answer more complex questions and give directions. The persistent tablet welcome screen (Figure 23) addressed the face detection reliability limitation identified by the Preusse et al. (2021) project. The tablet also served as an alternative input method mitigating the ego-noise limitation identified by Li et al. (2024) meaning the system was still interactable in a loud room. GDPR compliance was maintained throughout with no biometric data leaving the laptop at any point. The questionnaire showed that 80% of participants would prefer using the robot over waiting for a human receptionist (Figure 34) validating the core argument of the project.

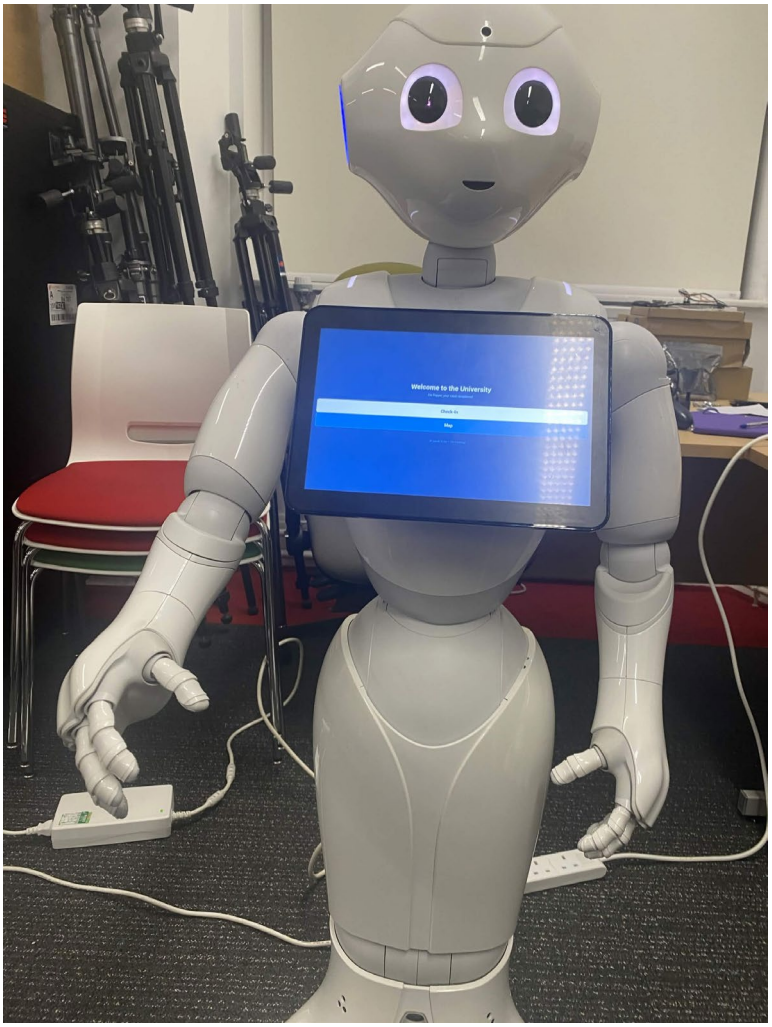


Figure 23 - Pepper showing persistent welcome screen on its tablet

5.2 Limitations

The system has several limitations identified during development and testing. A major limitation was that the system records for a fixed 5 second window rather than toggling based on sound detection. However, the laptop's microphone being used to test wasn't sensitive enough to detect between silence and speaking meaning silence could not be reliably distinguished from speech. There was no conversational memory between exchanges meaning that GPT-4o treated each exchange as a different conversation and couldn't remember what was said previously. Another issue was that Pepper occasionally moved its head during check-in causing the tablet to shift. The staff directory is hardcoded in app.py which wouldn't be scalable for real deployment as adding or removing staff would require code changes. A further limitation was that the GPT-4o response time caused delays with Pepper responding with 20% of participants finding it unacceptable (Figure 33). Face detection reliability still remains a limitation preventing a majority of the system from working as identified by Preusse et al. (2021). The system relies on the laptop microphone and camera rather than Pepper's built-in sensors which would be a portability issue for real deployment as it would be difficult to position the laptop. The study had limitations in that it was only a small sample size of five participants who were likely to be familiar with technology due to being university students which may not represent typical reception visitors such as the elderly or non-technical users. Furthermore, the testing was conducted in a closed laboratory environment eliminating external factors that would be present in real-world testing, and the testing was conducted in a single session meaning that long-term reliability is not guaranteed. A final limitation is that Aldebaran Robotics, the producer of Pepper, went into receivership meaning that Pepper may not be supported or produced in the future.

5.3 Reflection

The hardest part of this project was developing the Python bridge that allowed integration between Python 2.7 and Python 3 to allow the use of the modern AI libraries I needed to complete the project. This required many different iterations of code to get it working on Pepper simultaneously. I felt relieved when the bridge started working because I felt anxious being stuck on the initial steps of the project. If I started this project again, I would start creating functionality with Pepper earlier as I was discouraged early in the project because the Aldebaran Robotics website was frequently unresponsive making it difficult to access the SDK documentation. Furthermore, the scope of the project was overwhelming to begin with as the system required multiple components to be functional before any testing could be conducted requiring learning many new skills such as learning to code using the SDK and getting all the AI libraries to work to a good level. One thing that I learned that surprised me was that GPT-4o is surprisingly easy to work with and creates highly detailed responses from relatively little code. I am proud of what I built as I did not expect to successfully develop a fully integrated robotic system independently. I developed skills especially with NLP and integrating GPT-4o to give detailed informative responses about the university, AI project integration with libraries such as Whisper for speech transcription and OpenCV for facial detection, coding in Python for a large-scale project and robotics in terms of learning how to physically connect the robot and program it. Working by myself on this project was advantageous as it provided flexibility of the work to the challenges during development. However, solo development was challenging because it meant when I got stuck, I couldn't easily ask for help, it reduced accountability for meeting deadlines and increased the development time due to code not being created in parallel. An unexpected challenge was a network subnet mismatch between the university Wi-Fi and Pepper which prevented the laptop from communicating with the robot and it was unable to join due to a username requirement input not available with pepper. This was initially resolved by connecting via ethernet cable, before finding a more practical solution by sharing the laptop's mobile hotspot demonstrating the kind of unexpected hardware challenges that are difficult to anticipate in advance.

5.4 Future work

Future work that could be conducted in relation to this project would be true voice activity detection replacing the fixed 5 second recording window with detection that starts and stops based on speech directly addressing the timing issue participants experienced. A further improvement would be integrating interruption handling allowing the participant to interrupt Pepper while it is speaking to make the interaction feel more natural and conversational. Conversational memory would be integrated into the system by passing message history to GPT-4o so that the robot remembers what was said earlier in the conversation, addressing the F10 limitation and feedback from testing. A dynamic staff directory would be integrated into the code to allow for staff to be added or removed without code changes through replacing the hardcoded staff list with a database, and the email would be done through the university network rather than through Gmail to allow for practical results. The system could be improved by using Pepper's built-in microphone and camera rather than laptop sensors to make the system more portable and practical. Response time could be improved through different methods such as by using a faster model like GPT-4o-mini to reduce delays.

Features that could be implemented would be calendar integration to allow functionality such as booking a meeting in a calendar with a staff member through check-in however this was descoped due to time constraints. A date and time display was requested by a participant to be displayed on the tablet to be useful at a glance for people visiting such as knowing how long it is till their meeting. Integrating more physical interactions with Pepper such as being able to shake its hand was suggested and many participants were intrigued by this however this may not be feasible given the fragile nature of Pepper's joints. Multi-language support would also be a good addition to Pepper as it would help increase accessibility to international visitors and is supported by Pepper and Whisper.

Future deployment could include larger scale user testing with more diverse participants such as the elderly and non-technical users. It could be tested in a real reception environment outside the laboratory to allow for realistic testing conditions in real-world deployment.

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7 Appendices

7.1 Appendix 1: SDLC details

CS3IP Project Definition Form

Student Name/ID: 230163795
Course: Computer Science
Project title: Robot Receptionist
Supervisor: Ulysess Bernadet
Date: 9/10/25

Project overview:
This project aims to program a robot receptionist that is emotive with greetings and can handle basic tasks.

Project deliverable:
- Speech to text functionality (can listen to clients)
- Machine vision/image recognition
- Ask and remember client names
- Access a defined calendar and update it based on information it is told
- Multiple different emotive and exciting greetings (such as running round in a circle when meeting someone)
- Path finding across a room
- Answer questions and give directions

What is interesting & useful about this project? Can you make a business case for it?

The project will tackle the issues of human receptionists such as being available to work 24/7 365 days a year. It wouldn't need time off or breaks and would only require short times for maintenance and charging. A robot doesn't get tired like humans, and it would consistently be able to complete tasks such as automatically scheduling appointments straight into the system. It would invariably be able to give enthusiastic greetings no matter the time of the day meaning that it can maintain a positive atmosphere around the reception. The robot would give novelty to the business and could impress clients as they do not often see them. Moreover, it may be more approachable than a human receptionist reducing the chances of social anxiety or embarrassment when asking for information such as directions as shown by different robots such as order terminals.

Outline Project Timetable:

No.	Start Date	End Date	Task
1	21/10/2025	19/04/2026	Writing report
2	21/10/2025	22/10/2025	Research coding for robot
3	30/10/2025	04/11/2025	Research similar projects
4	30/10/2025	06/11/2025	Decide project requirements
5	21/11/2025	21/12/2025	Create prototype
6	21/10/2025	20/11/2025	Initial testing
7	21/10/2025	05/11/2025	Debugging and improvments
8	01/12/2025	01/03/2026	Editing report
9	01/04/2025	11/04/2025	Preparing for presentation
10	21/04/2025	22/04/2025	Submit report

Figure 24 – The original Project definition form submitted in October, outlining the initial project scope and timetable

Planned	
Actual	

Table 4 - Project Gantt chart showing planned versus actual timeline

Task names	October	November	December	January	February	March	April
Literature review (Planned)							
Literature review (Actual)							
Requirements specification (Planned)							
Requirements specification (Actual)							
Flask web server (Planned)							
Flask web server (Actual)							
Speech recognition (Planned)							
Speech recognition (Actual)							

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GPT-4o integration (Planned)							
GPT-4o integration (Actual)							
Face detection (Planned)							
Face detection (Actual)							
Bridge architecture (Planned)							
Bridge architecture (Actual)							
Check-in system (Planned)							
Check-in system (Actual)							
Testing and evaluation (Planned)							
Testing and evaluation (Actual)							
Report writing (Planned)							
Report writing (Actual)							

7.2 Appendix 2: Ethics

The full ethic files are submitted in additional materials in the folder ethics.

Usability Test Tasks

Task 1: Saying Hello Go up to the robot and see if it greets you and tells you its name

Task 2: Finding Your Way Imagine you have a meeting on an upper floor. Ask the robot for the best way to get there, such as where the lift or the stairs are located.

Task 3: Getting Information Ask the robot a specific question about the building.

Task 4: A Harder Question Try asking the robot something a bit more detailed. For instance, ask it to recommend a quiet place to work nearby.

Task 5: Checking into the building Try asking the robot to check you into the building.

Figure 25 - Usability test tasks

Aston University
BIRMINGHAM UK

Robot Receptionist

Consent Form

Name of Chief Investigator: Oliver Blatchford

		Please initial boxes
1	I confirm that I have read and understand the above study information. I have had the opportunity to consider the information, ask questions and have had these answered satisfactorily.	
2	I understand that my participation is voluntary and that I am free to withdraw at any time up until the point I submit my responses, without giving any reason and without my legal rights being affected.	
3	I agree to my personal data and data relating to me collected during the study being processed as described above.	
4	I agree to take part in this study.	
5	I confirm that I have read and understand the above study information. I have had the opportunity to consider the information, ask questions and have had these answered satisfactorily.	

Figure 26 - Consent form for the testing

7.3 Appendix 3: Questionnaire results

The questionnaire questions can be accessed via this link: https://forms.gle/1aXY4rGW12JgLCXm6

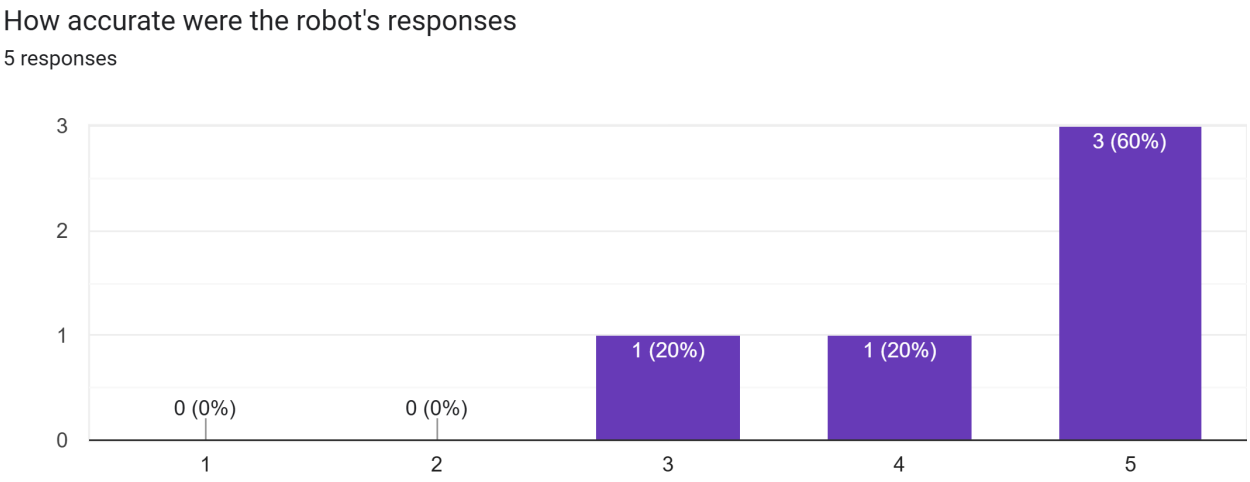


Figure 27 - How accurate were the robot's responses

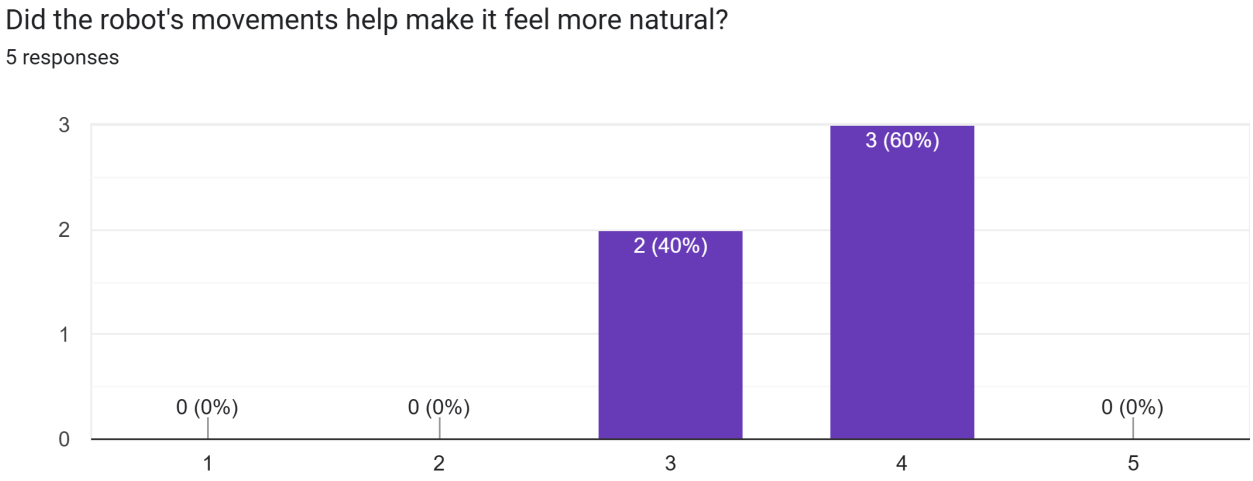


Figure 28 - Did the robot's movements help make it feel more natural?

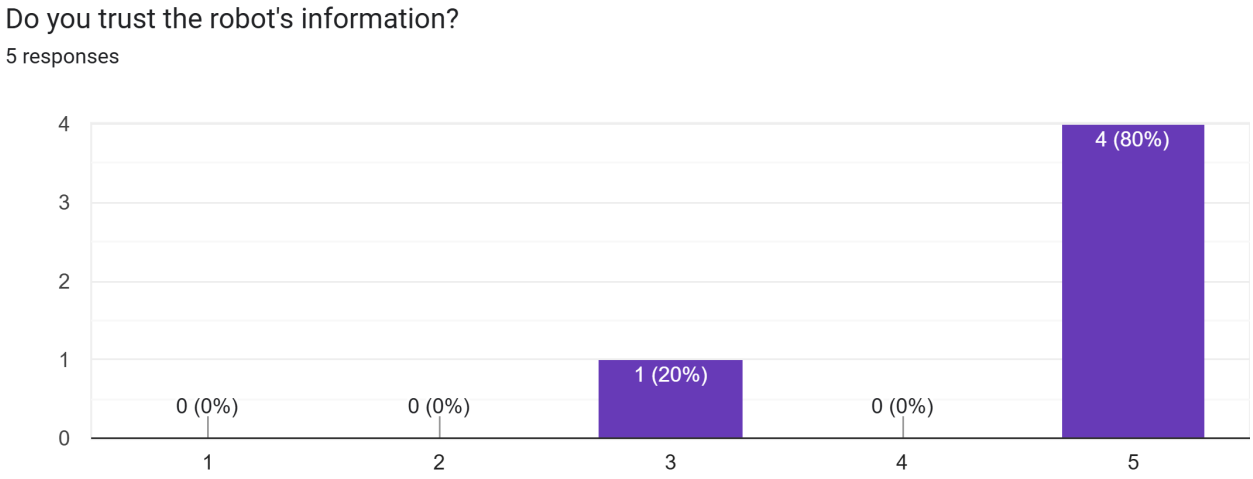


Figure 29 - Do you trust the robot's information?

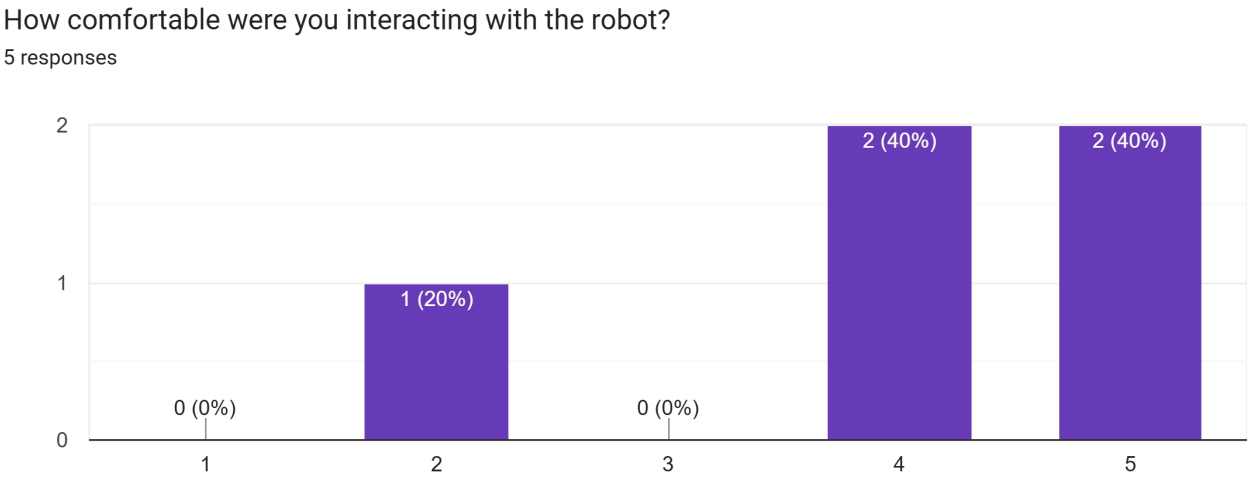


Figure 30 - How comfortable were you interacting with the robot?

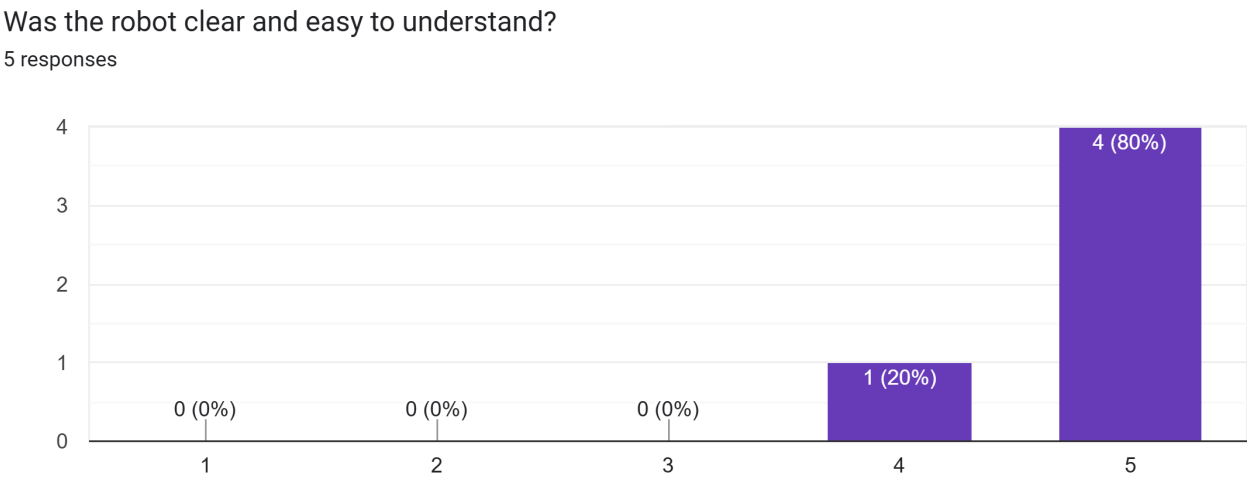


Figure 31 - Was the robot clear and easy to understand?

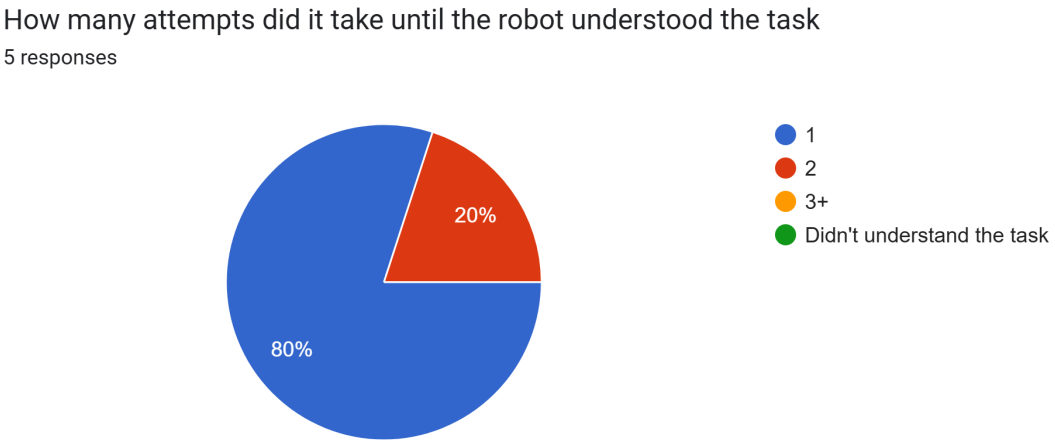


Figure 32 - How many attempts did it take until the robot understood the task

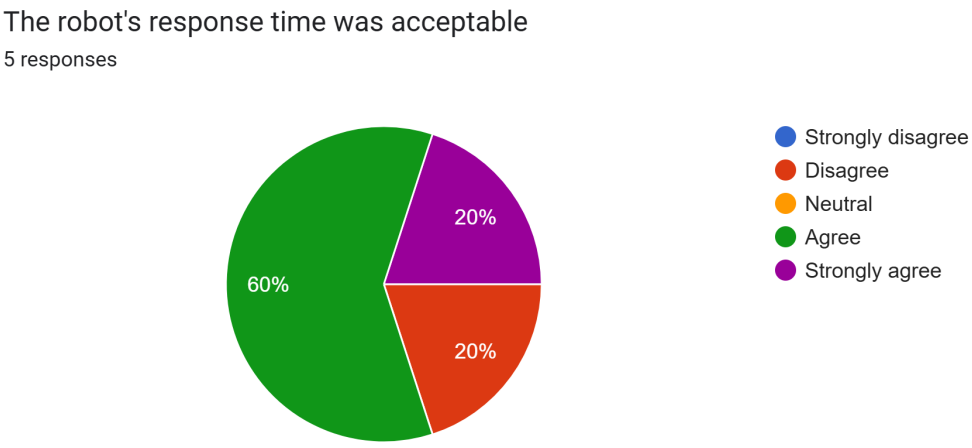


Figure 33 - The robot's response time was acceptable

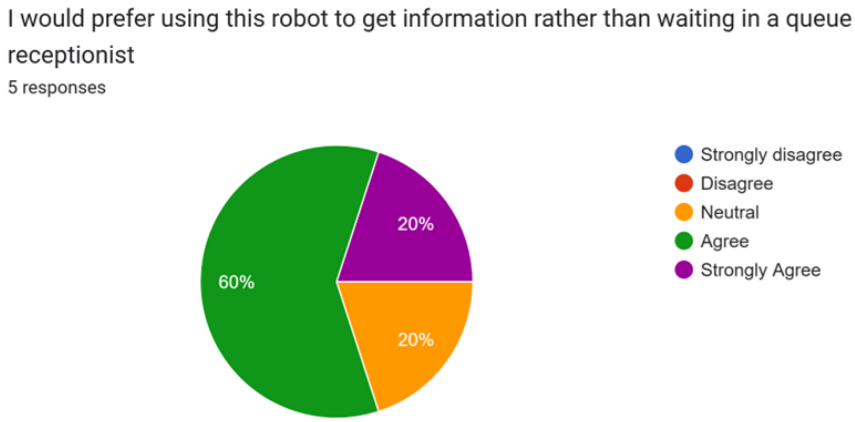


Figure 34 - I would prefer using this robot to get information rather than waiting in the queue for a human

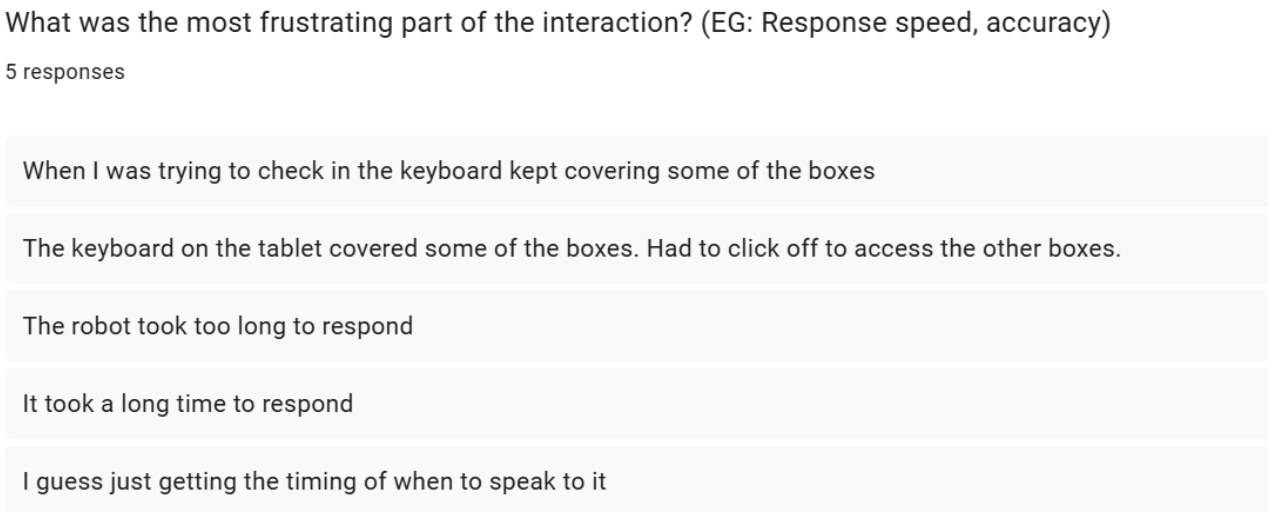


Figure 35 - What was the most frustrating part of the interaction? (EG: Response speed, accuracy)

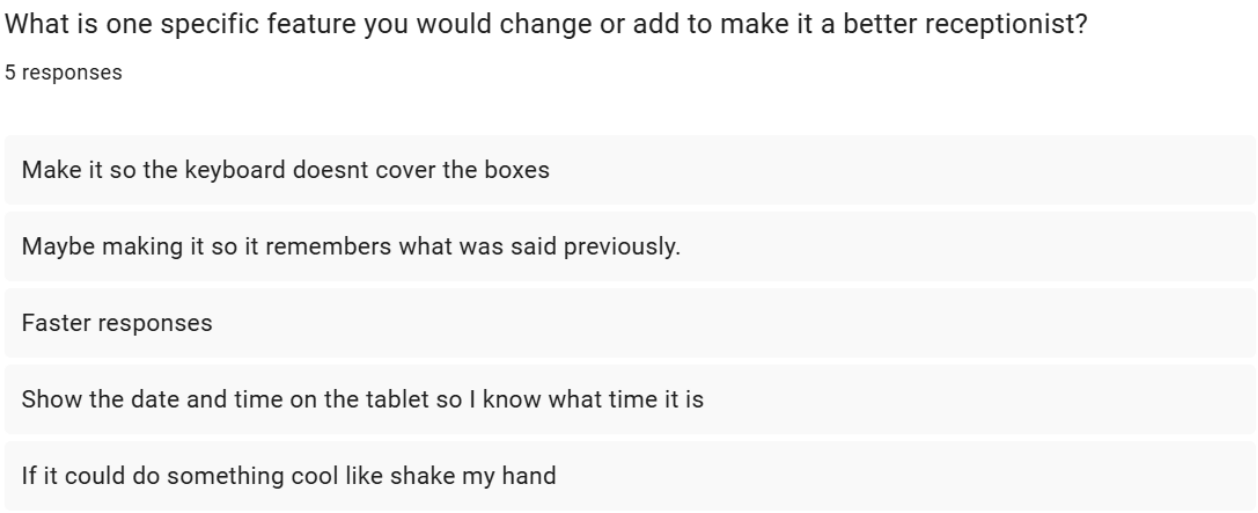


Figure 36 - What is one specific feature you would change or add to make it a better receptionist?

7.4 Appendix 4: Meetings with supervisor

Table 5 - Meetings with supervisor with date and description of what occurred

Date	Discussed
2/10/2025	Initial project meeting – discussed project scope, Pepper robot capabilities and initial objectives
9/10/2025	Refining scope of project and requirements
16/10/2025	Checking the project definition form and agreeing on the plan for the project
16/10/2025 to 27/2/2026	Independent development period. Iterative feature-by-feature implementation following Agile methodology
27/2/2026	Showing report progress and getting feedback on sections and asking for help with Pepper's connectivity
6/3/2026	Discussed development progress with supervisor, reported on the system functionality and outstanding implementation challenges
13/3/2026	Showing ethics documentation progress and getting feedback
20/3/2026	Submitting ethics documentation and getting signature
21/4/2026	Showing report progress and getting feedback on sections

7.5 Appendix 5: Code

The full source code is submitted in additional materials in the folder PepperReceptionist.